

SATHANA S

Aspiring Data Scientist & Machine Learning Engineer | Salem, Tamil Nadu, India
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PROFESSIONAL SUMMARY

B.Sc. Artificial Intelligence & Data Science student pursuing a career as a Data Scientist / Machine Learning Engineer, with hands-on internship experience in machine learning, EDA, classification modeling, and predictive analytics. Proficient in Python, SQL, Scikit-learn, XGBoost, and Random Forest, skilled in end-to-end ML pipelines from preprocessing and feature engineering to model deployment via Streamlit and Flask. Experienced with cross-validation, hyperparameter tuning, TF-IDF vectorization, data visualization, and Agile development workflows.

Core Competencies: Machine Learning · Artificial Intelligence · Data Analytics · Predictive Modeling · Data Visualization · Statistical Analysis · Model Deployment · Natural Language Processing (TF-IDF)

Soft Skills: Problem-Solving · Analytical Thinking · Communication · Team Collaboration · Agile Workflow · Time Management

TECHNICAL SKILLS

Languages	Python, SQL
ML / AI	Scikit-learn, XGBoost, Random Forest, Logistic Regression, Decision Tree, Collaborative & Content-Based Filtering
Data Science	EDA, Feature Engineering, Predictive Modeling, Classification, Regression, Hyperparameter Tuning, Cross-Validation, TF-IDF, Cosine Similarity, Model Evaluation, Data Wrangling
Data Tools	Pandas, NumPy, Matplotlib, Seaborn
Deployment	Streamlit, Flask, REST API Design, Microservices (basic)
Platforms	Jupyter Notebook, Google Colab, MySQL, Git, GitHub, MS Excel

PROFESSIONAL EXPERIENCE

Data Science Intern Mar 2026 – May 2026
InternsElite

- Cleaned and preprocessed 10,000+ record datasets, reducing data inconsistencies by ~35% for downstream ML pipeline tasks.
- Conducted comprehensive EDA using Pandas and Seaborn, uncovering actionable patterns across customer, sales, and operational datasets.
- Developed and evaluated Logistic Regression, Decision Tree, and Random Forest classifiers, achieving up to 87% accuracy on holdout test sets.
- Applied cross-validation and hyperparameter tuning across Agile sprint cycles, reducing overfitting by ~15%.

Machine Learning Intern May 2026
LIVEWIRE

- Engineered a hybrid Movie Recommendation System using collaborative and content-based filtering, improving personalization accuracy by 20% over baseline.
- Applied TF-IDF vectorization and cosine similarity across a 5,000+ title dataset, exposing results via a REST API design pattern.
- Evaluated model performance with precision and recall metrics; delivered top-5 suggestions with 82% user relevance.

PROJECTS

Road Accident Severity Predictor [\[GitHub\]](#) — Python · Random Forest · XGBoost · Streamlit

- Built a multi-class classifier (Fatal/Grievous/Minor/No Injury) on 600+ Tamil Nadu accident records (2018–2024); XGBoost achieved 89% accuracy, 0.87 weighted F1-score.
- Engineered domain features (Speed_Excess, Time_of_Day_Risk, Weather_Severity_Index), boosting F1 by 12%; deployed as a real-time Streamlit app.

Movie Recommendation System [\[GitHub\]](#) — Python · TF-IDF · Cosine Similarity

- Built a hybrid recommender combining collaborative and content-based filtering on a 5,000+ title dataset.
- Applied TF-IDF vectorization and cosine similarity to generate top-5 personalized suggestions with high relevance.

Customer Churn Prediction [\[GitHub\]](#) — Python · Scikit-learn · EDA

- Built a classification pipeline to predict customer churn; performed EDA, feature engineering, and model evaluation.
- Compared multiple classifiers with cross-validation to identify the best-performing churn prediction model.

Diabetes Risk Prediction [\[GitHub\]](#) — Python · Random Forest · Feature Engineering

- Built a healthcare classifier on the 768-record PIMA dataset (8 health metrics), achieving 83% accuracy.
- Applied IQR outlier handling and Random Forest hyperparameter tuning to improve model robustness.

EDUCATION

B.Sc. in Artificial Intelligence & Data Science 2024 – 2027
Padmavani Arts & Science College, Salem, Tamil Nadu

CERTIFICATIONS

- Data Visualisation: Empowering Business with Effective Insights — Forage / Tata (July 2026)
- Data Analytics Job Simulation — Deloitte (June 2026)
- Data Science Internship Certificate — InternsElite (May 2026)
- Data Science Industrial Training Certificate — InternsElite
- Machine Learning Based Movie Recommendation System Internship — LIVEWIRE (May 2026)